

Flash Twenty LEDs Using Arduino

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Presented here is the circuit that drives twenty LEDs in a random manner. LED flashers and sequencers are used in decorative lights to give pretty colour combinations by glowing LEDs in random or sequential manner. This flasher has a chip that controls the random flashing rate of the LEDs.

Circuit and working

The circuit diagram of the LED flasher using Arduino is shown in Fig. 1. The circuit is built around Arduino Nano board (Board1), twenty coloured LEDs (LED1 to LED20) and a few other components.

Arduino is programmed in such a way that only one LED glows at a time. A random number is generated by the program that is used to turn on the LED connected to the particular I/O pin of Arduino. On delay time period between any two LEDs or next random

LED is controlled using potmeter VR1. Thus, all LEDs flash in a random manner by illuminating one LED at a time. These glow in this pattern continuously until power is switched off or removed.

Arduino Nano. Arduino Nano is a compact and breadboard-friendly board based on ATmega328 microcontroller (MCU). It has similar function-

ality as Arduino Uno but is available in a different and smaller package.

Dimensions of Arduino Nano are 43mm x 18mm. It has fourteen digital I/O [of which six are pulse-width modulation (PWM) I/O pins and eight analogue inputs], 16MHz clock speed and 32kB flash memory. It uses a mini USB cable instead of the standard USB one for connecting to the computer.

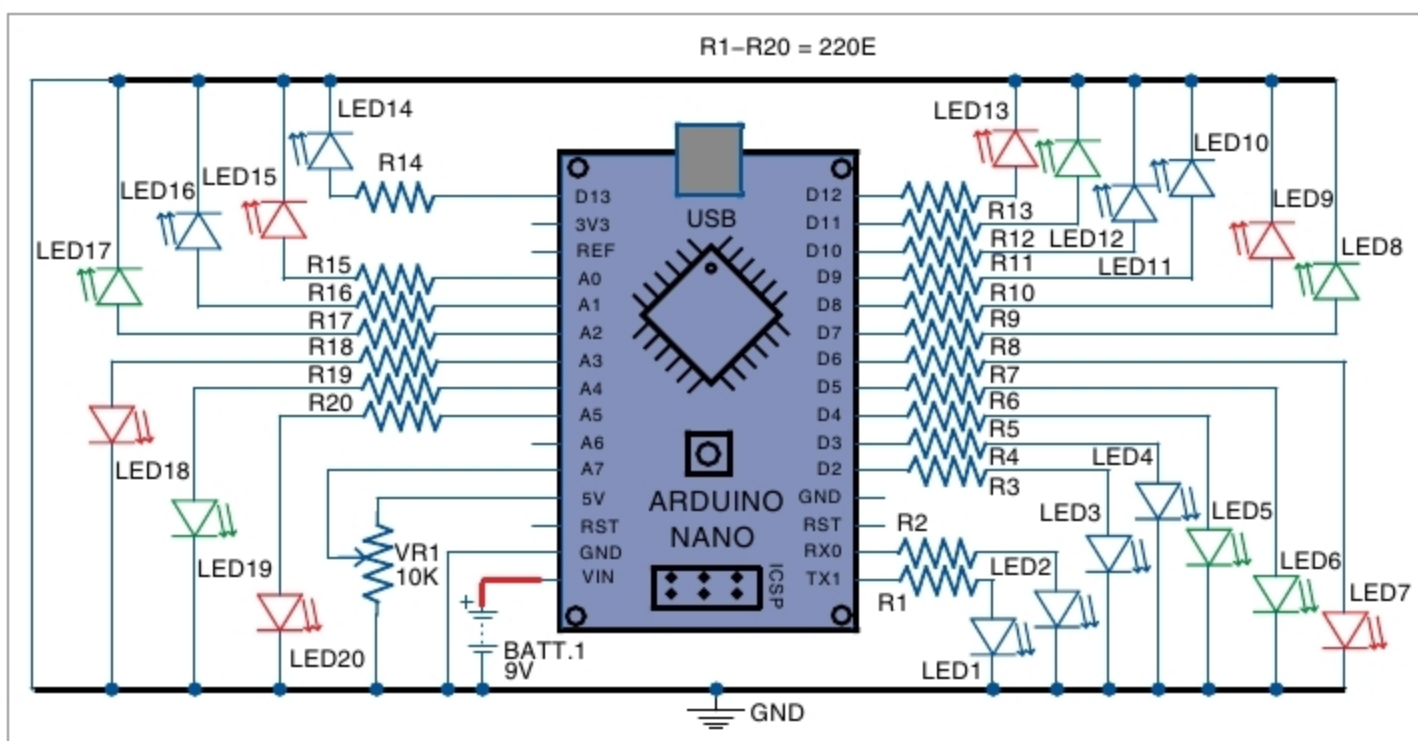


Fig. 1: Circuit diagram of the LED flasher using Arduino

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